InsightVLM-20K: chart images paired with statistically verified findings

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Abstract

Charts carry most quantitative findings to readers, yet no public chart corpus pairs an image with a statement about the plotted data that has been checked to be true. InsightVLM-20K contains 20,000 charts in 26 chart types, each labelled with one of 20 statistical findings such as bimodality, heteroscedasticity or a level shift. A label is issued only when a named statistical test confirms it on exactly the plotted array and no other class matches; re-running the tests reproduces 99.95% of labels. Each sample carries its data table, statistics, a reasoning chain, a numerically grounded insight and evidence boxes. 6,083 charts come from 654 real public tables in source-disjoint splits, so generalisation to real data is measured directly. We document a shortcut audit, a rendering leak we found and removed, and test sets re-drawn by two other plotting libraries. A fine-tuned CNN ensemble reaches 0.867 on real test charts against 0.468 for the best zero-shot vision–language model, and 0.716 on charts from an unseen plotting library.

Background & Summary

Quantitative evidence is usually met as a picture. A reader sees a distribution, a correlation or a trend long before the numbers behind it, and the properties that most often invalidate a naive reading, such as two modes behind one mean, variance that grows with the predictor, or a visible gap between groups that is within sampling noise, are visible in the figure but rarely stated in its caption.

Chart understanding datasets are mature, but their supervision describes or queries the chart rather than stating what the data shows. FigureQA[1], DVQA[2], PlotQA[3] and ChartQA[4] pair charts with questions and answers; ChartGen[5] and ChartCoder[6] pair them with plotting code; ChartQA-X[7] adds explanations of answers; and reasoning benchmarks such as CharXiv[8] test description and reasoning over scientific figures. None attaches a label that a statistical test has confirmed on the plotted data, so none can be used to train or score a model on whether a stated finding is true.

InsightVLM-20K fills that gap. Its design goals were: (i) verified labels: a label exists only if a named test confirms it on the exact plotted array, and only if exactly one class matches; (ii) balance: exactly 1,000 samples per class, split 700/100/200; (iii) measurable generalisation: real-source charts sit in source-disjoint splits and are reported separately from synthetic ones; (iv) auditable shortcuts: every known way to predict the label without reading the data is measured, and the dataset's own rendering is tested for leaks.

The dataset supports supervised chart classification, vision–language instruction tuning (instruction/response pairs are included), chart-to-table reconstruction, and evidence grounding. Because every sample stores its data table and statistics, it also supports studies of how well models' cited numbers match the data.

Table 1: The 20 finding classes and the tests that license them.

Family	Class	Verification guard
Distribution	dist_normal	Shapiro–Wilk $p > 0.05$, $\lvert\text{skew}\rvert < 0.35$, dip test[9] $p > 0.15$
	dist_right_skew	skew > 1.0 and Bowley skew > 0.05
	dist_left_skew	skew < -1.0 and Bowley skew < -0.05
	dist_bimodal	dip $p < 0.05$, GMM BIC gain > 10, KS-uniform $p < 0.05$
	dist_heavy_tail	excess kurtosis > 2.0, outliers $> 2.5\%$, $\lvert\text{Bowley}\rvert < 0.10$
	dist_uniform	KS vs. uniform $p > 0.10$, excess kurtosis < -0.9
Relationship	rel_strong_positive	Pearson $r > 0.70$, $\lvert r - \rho\rvert < 0.09$
	rel_strong_negative	$r < -0.70$, $\lvert r - \rho\rvert < 0.09$
	rel_none	$\lvert r\rvert < 0.12$, $\lvert\rho\rvert < 0.12$, quadratic $R^2 < 0.05$
	rel_nonlinear_monotonic	$\lvert\rho\rvert > 0.80$, $\lvert\rho\rvert - \lvert r\rvert > 0.10$, linear $R^2 < 0.72$
	rel_nonlinear_nonmono.	$\lvert r\rvert < 0.25$, quadratic $R^2 > 0.45$, $\lvert t_{\text{curv}}\rvert > 4$
	rel_heteroscedastic	Breusch–Pagan[10] $p < 0.001$, spread ratio > 2.5
Temporal	ts_increasing	Mann–Kendall[11] $p < 0.01$, $\tau > 0.35$
	ts_decreasing	Mann–Kendall $p < 0.01$, $\tau < -0.35$
	ts_seasonal	periodogram power > 0.30, ACF > 0.35, $\lvert\tau\rvert < 0.25$
	ts_changepoint	SNR > 2.5, step model beats line, abruptness > 0.55
	ts_anomaly_spike	robust $\lvert z\rvert > 6$, no trend after removal
Group	grp_significant_diff	Kruskal–Wallis[12] $p < 0.001$, $\varepsilon^2 > 0.14$
	grp_no_diff	Kruskal–Wallis $p > 0.30$, $\varepsilon^2 < 0.02$
	grp_dominant_category	top share > 0.55, normalised entropy < 0.75

Methods

Finding taxonomy

Twenty finding classes span four families an analyst routinely reports (Table 1): six distribution shapes, six relationship types between two variables, five time-series behaviours and three group or composition findings. Each class is (i) verifiable by a standard test, (ii) mutually exclusive with the others by explicit guards, and (iii) visible in at least four chart types.

Verification gate

Three rules govern labelling. Subsample before verifying: data are reduced to their final plotted size before any test runs, so the label describes what the image shows rather than a parent table. Exactly one class: a candidate satisfying two classes' guards is discarded rather than multi-labelled. Reject rather than mislabel: every guard is conservative; more candidates can always be mined, a wrong label cannot be recovered. Robust statistics are used where naive ones fail in measurable ways: moment skewness alone labelled 12.5% of symmetric Student-$t(4)$ samples as skewed, so skew classes also require Bowley skewness; the dip test is calibrated against the uniform, so bimodality also requires a KS test against the uniform to reject; and a linear ramp produces a high-SNR "changepoint", so level shifts also require a two-level model to beat a line and an abruptness ratio above 0.55. Over 10 seeds $\times$ 20 classes, generators produce their intended class 95.5% of the time; every miss is a rejection, not a wrong label.

Source data

Real track. We harvested 719 public tables from OpenML[13], the World Bank, Our World in Data, vega-datasets, seaborn-data and UCI, and mined every plottable view: single columns, column pairs, per-entity time series, numeric-by-category groupings and categorical compositions. Real data cannot fill every class: across all 719 tables only 10 clean inverted-U relationships exist. Synthetic track. Data whose property holds by construction were generated with domain vocabularies (health, economics, environment, engineering, education, retail) and realistic names, units and magnitudes. Both tracks pass the identical gate. Provenance is balanced within each class rather than filled real-first, so that no class is dominated by one provenance. The final corpus has 6,083 real samples from 654 source tables (30.4%) and 13,917 synthetic samples.

Splits

Samples are split 14,000/2,000/4,000 (700/100/200 per class). Real samples are split by source table: every chart derived from one table is in one split. The real subsets contain 4,338 training, 553 validation and 1,192 test charts; real class frequencies are uneven because public data are (Supplementary Table S1).

Supplementary real training pool

The balanced set caps real data at 500 samples per class, which leaves verified real candidates unused. We release them as an optional training pool: candidates whose source table the split hash assigns to train, with exact duplicates of any dataset sample removed and at most 400 per class. By construction no validation or test source table contributes to it (the build asserts this). It adds 2,545 real training charts in nine classes (Supplementary Table S1), raising real training data from 4,338 to 6,883 charts. It is kept separate from the balanced splits.

Rendering

Charts are drawn with matplotlib[14] and seaborn[15] on a fixed 768×576 canvas in 26 chart types and 8 visual styles. The style of each sample is chosen from a CRC32 hash of its identifier, so rendering is reproducible across processes. Tight bounding-box cropping is disabled, because it resizes the canvas after the axes transform is read and would shift every evidence box. Dates stored as epoch timestamps are plotted as calendar dates.

Label-dependent marks were removed (version 2). The first release drew annotations on time-series charts that depended on the label: a fitted trend line only on increasing and decreasing series, a vertical line at the change point only on level-shift series, and circles only around anomalies. These marks reveal the class without the data being read, and trained models used them (see Technical Validation). Version 2 draws no label-dependent marks; the corresponding evidence is kept only as grounding boxes in the metadata. No other chart family drew label-dependent marks (regression lines on scatter plots are drawn independently of the label). Figure 1 shows both renders of the same data.

Cross-renderer test sets

To test whether models learn matplotlib's appearance rather than the statistics, all 1,192 real test tables were re-drawn with two further libraries. Plotly[16] differs in fonts, gridlines, colours, marker shapes, binning rules and layout; five templates are assigned by hash, excluding the one that imitates seaborn. Vega-Lite (rendered with vl-convert) uses nine themes, excluding those



Figure 1: The same real data rendered with the version-1 label-dependent mark (left) and in the released version-2 form (right). Top: a level shift in French unemployment; bottom: anomalies in a mobility series.

that match a matplotlib training style. Chart types map to each library's nearest native form; where none exists (for example boxen or hexbin plots) the substitute is a form the taxonomy lists as valid for every class that chart type can carry. No label-dependent marks are drawn.

Training-only renderer augmentation

For training we additionally release two alternative renders of every training and supplementary chart: a Plotly render, and a matplotlib render with randomised fonts, font sizes, line and marker sizes, colours, background tints, gridlines, spines and title placement (data and chart type unchanged). No validation or test chart is rendered this way, and no Vega-Lite render is used for training, so Vega-Lite remains a renderer that no model has seen.

Additional annotations

Each sample stores the plotted data table, a statistics dictionary (moments, quantiles, correlation coefficients, test statistics, p-values and effect sizes), the name of the test that licensed its label, a reasoning chain (observation → statistic → test → conclusion), an insight paragraph in which every claim cites a computed number, and pixel boxes for the plot area and the supporting evidence (for example the modal bin, the outlier band, the change point or the dominant wedge).

Table 2: Files in the release.

Path	Contents
images/<class>/<id>.png	20,000 chart images, 768 × 576 RGBA PNG
records.jsonl	one JSON record per sample: identifiers, label, family, chart type, split, provenance, source and licence, data table, statistics, test, reasoning chain, insight, render metadata and evidence boxes
tables/metadata.parquet	the same fields flattened for analysis
splits/{train,val,test}.jsonl	records per split
splits/test_real_only.jsonl	the 1,192 real-source test samples
instruct/{split}.jsonl	image/prompt/response triples for instruction tuning
test_renders/{plotly,vega}/	Plotly and Vega-Lite re-renders of the 1,192 real test charts
train_extra/	supplementary real training pool: 2,545 images and records_extra.jsonl
train_aug/{plotly,mplrand}/	alternative renders of every training and supplementary chart
class_index.json	label ↔ id map, class descriptions, valid chart types, tests
DATASHEET.md, BIAS_AUDIT.md	datasheet[17] and audit report

Data Records

The dataset is deposited at [Zenodo/Figshare DOI] under CC BY 4.0 and mirrored on Kaggle (https://www.kaggle.com/datasets/asifxzaman/insightvlm-20k-v2). Table 2 lists the files.

Sample identifiers have the form <class>___<index>. Provenance is real or synthetic; real samples record their source repository, source table identifier, URL and licence. Difficulty (easy/medium/hard) is the within-class tertile of the evidence margin, i.e. how far the deciding statistic lies beyond its guard.

Technical Validation

Label correctness and integrity

Re-running the verification battery on each test sample's stored data table reproduces 3,998 of 4,000 labels (0.9995); the two exceptions lie within 1% of the heteroscedasticity guard and fail to no class rather than to a wrong class under a different statsmodels build. The integrity audit finds 0 of 654 source tables in more than one split, 0 duplicate data fingerprints and one visual duplicate among 20,000 images. A near-duplicate check (comparing every real validation and test chart with every trainable real chart of the same arity: signatures correlating above 0.99, then the same values, rows or series at matching scale) flags 85 real test charts (7.1%) whose data also appears, under a different source table, in training data; most are republished tables (for example World Bank and Our World in Data internet-use series, or two uploads of the same OpenML dataset). Removing them changes the ensemble's real-chart accuracy from 0.867 to 0.863 (95% CI $[0.843, 0.883]$); their identifiers are released so results can be reported on the deduplicated set.

Shortcut audit

A shortcut is any feature that predicts the label without the statistical evidence. Eleven were measured on version 2 against thresholds fixed in advance, and all pass (Table 3). Real and synthetic charts are not separable from low-level pixel statistics (AUC 0.551), so the synthetic majority does not give provenance away. Chart type is associated with the label by construction (a pie chart can only carry a composition finding); we disclose this and use it as the evaluation floor rather than removing it.

Table 3: Shortcut audit on version 2. All eleven tests pass.

Test	Value	Threshold
Class balance (imbalance ratio)	1.000	$= 1.000$
Split balance (max s.d. of class proportions)	0.0000	< 0.01
Chart type $\rightarrow$ label (prior accuracy)	0.2883	reported as floor
Render style $\rightarrow$ label (Cramér's V)	0.0000	< 0.05
Provenance detectability from pixels (AUC)	0.551	< 0.75
Axis text $\rightarrow$ label (macro-F1)	0.3688	< 0.55
Pixel statistics $\rightarrow$ label (macro-F1)	0.4038	< 0.55
Source concentration (top share in a class)	17.2%	$< 35\%$
Length confound within family (ratio)	2.04	< 4.0
Subject-domain concentration	22.8%	$< 45\%$
Category-vocabulary concentration	13.0%	$< 35\%$

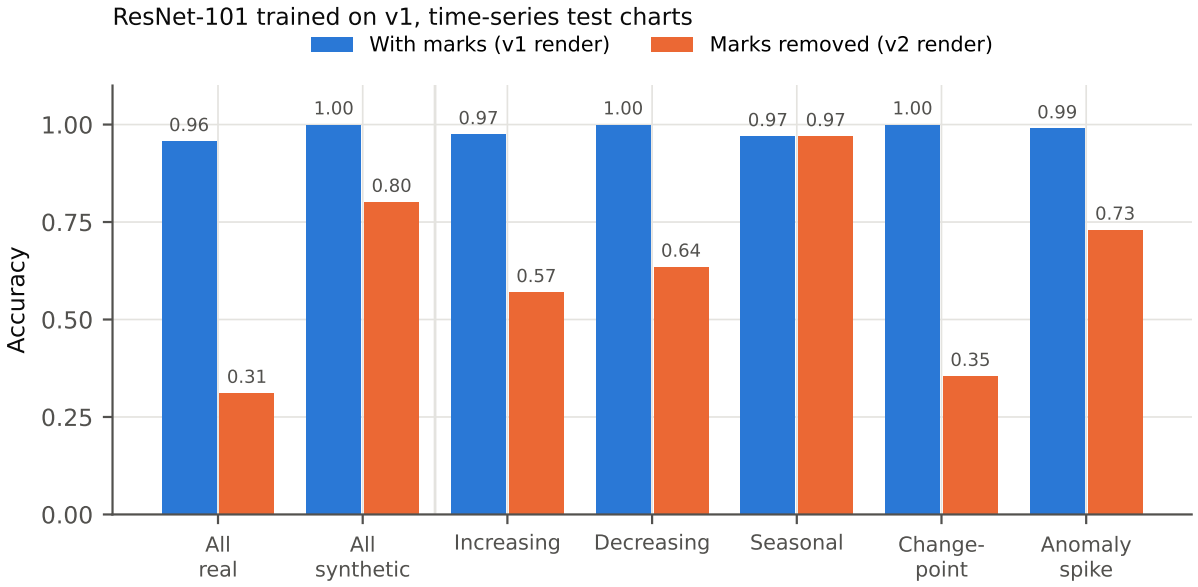


Figure 2: Accuracy of a ResNet-101 trained on version 1 on test time series rendered with and without the label-dependent marks. Only the marks differ between the two renders.

A rendering leak, measured and removed

The shortcut audit examines metadata and pixel statistics, not what a trained model attends to, and it did not detect the version-1 time-series marks. We found them by inspecting the renderer and measured their effect directly: the same 1,000 test time series were rendered twice with identical styles, once with the marks and once without, and scored with two ResNets trained on version 1 (Figure 2). With marks, accuracy on real time series was 0.957 (ResNet-101) and 0.944 (ResNet-50); without them it fell to 0.311 and 0.407, with the level-shift class dropping from 1.000 to 0.355. Seasonal series, the one temporal class that carried no mark, were unaffected (0.97 in both renders). Zero-shot vision–language models, which never saw version-1 training data, changed by only 1–2.5 points on all real charts and 4–10 points on real time series. This is shortcut learning[18] in its plainest form, and we report it because version-1 results in earlier drafts of this work overstated real-data accuracy. All results below use version 2.

Evidence grounding

A Grad-CAM pointing game on the 1,023 real test charts that carry evidence boxes checks whether the deployed models attend to the evidence that licenses each label. The attention peak falls inside an evidence box on 52.4% of charts against 43.9% for a random point in the plot area (gain +8.5 points, 95% CI $[+6.8, +10.2]$): clearly for time series (+15.4) and group comparisons (+11.0), less for relationships (+7.3), and not for distributions (−0.2). The evidence boxes are therefore usable as a grounding reference for three of the four families.

Evaluation protocol

With 20 balanced classes chance is 5%, but a rule that uses no pixels, predicting the most common class for each chart type, reaches 0.288; we use it as the floor. A data oracle that runs the verification battery on the stored table reaches 0.9995 and is the ceiling. The headline metric is accuracy on the 1,192 real test charts, with 95% bootstrap confidence intervals[19]; paired differences between systems are tested with exact McNemar tests[20] on the same charts. Mixed-test accuracy is reported for completeness but is dominated by synthetic charts.

Baselines

Fine-tuned CNNs. ResNet-101 and ResNet-50[21] and DenseNet-121[22], ImageNet-initialised, were trained at 384×384 for up to 12 epochs (AdamW, one-cycle schedule, label smoothing 0.1). Horizontal and vertical flips are never used: mirroring converts a positive correlation into a negative one and an increasing series into a decreasing one. Batches are sampled so that about half are real charts, and the checkpoint is chosen on real validation accuracy. Members are trained on the balanced training split, on the split plus the supplementary pool, and on the latter with renderer augmentation (each epoch shows each training chart as one of its three renders, uniformly). The deployed ensemble is the equal-weight average of the models trained with the full recipe (supplementary pool and renderer augmentation); only a label-shift correction[23] towards real-data class frequencies is tuned, on validation. Membership was set after all models were trained, so we also report the alternatives: averaging every trained model gives 0.862 on real test charts, and greedy ensemble selection[24] on validation gives 0.865. With 553 real validation charts, greedy selection chases one- to two-point differences that do not reliably transfer.

Zero-shot vision–language models. Qwen2.5-VL-3B and 7B[25] and Qwen3-VL-8B[26] received each chart with all 20 class names and one-line definitions and were asked for one class name, with greedy decoding. All three produced a parseable class for more than 99.9% of charts. They were evaluated on the matplotlib renders only.

Human validation. In a label check, 188 real validation and test charts (up to 10 per class) were shown with the finding their label asserts, mixed with 25 decoys showing a wrong finding from the same family; annotators answered yes, no or cannot tell from the chart alone, blind to each other and to the statistics. Annotators accepting more than 40% of decoys were to be excluded (rule fixed in advance; none was). Two annotators, one an author, confirmed 73.7% of labels (95% CI 67.6–79.0%), rejected 16.8% and could not tell for 9.6%; they rejected 78% of decoys and accepted 4%, with Fleiss' $\kappa = 0.68$ between them and 0.84–0.85 agreement with their own repeat pass. Confirmation was 0.95–1.00 for clear-cut findings (trends, normal and uniform distributions, dominant category) but 0.25 for bimodal and heteroscedastic and 0.40 for monotonic non-linear and anomaly-spike charts: those labels are statistically correct but the finding is subtle to the eye, which users should keep in mind when treating the labels as what a reader would perceive.

Table 4: Accuracy on the 1,192 real test charts drawn by three renderers, and on synthetic test charts. Brackets: 95% bootstrap CI. †Fixed 400-chart synthetic sample.

	Real test charts			
System	matplotlib	Plotly	Vega-Lite	Synthetic
Chart-type prior (no pixels)	0.288 (all test charts)			
Qwen2.5-VL-3B, zero-shot	0.306 [0.281, 0.332]	–	–	0.303†
Qwen2.5-VL-7B, zero-shot	0.411 [0.383, 0.440]	–	–	0.498†
Qwen3-VL-8B, zero-shot	0.468 [0.439, 0.496]	–	–	0.648†
ResNet-101, balanced split	0.841	0.747	0.607	0.997
+ supplementary real pool	0.840	0.735	0.602	0.999
+ renderer augmentation	0.859	0.854	0.650	0.998
DenseNet-121, + supplementary pool	0.861	0.771	0.638	0.997
+ renderer augmentation	0.855	0.852	0.712	0.998
Ensemble of the two full-recipe models	0.867 [0.846, 0.886]	0.857 [0.836, 0.877]	0.716 [0.690, 0.741]	0.999
Statistical oracle (ceiling)	0.9995 (all test charts)			

Table 4 and Figure 3 summarise the baselines. The deployed ensemble, the equal-weight average of the two full-recipe models, reaches 0.867 on real test charts and 0.999 on synthetic ones. Every zero-shot vision–language model is far below it (best 0.468; exact McNemar $p < 10^{-100}$ for each pairwise comparison), and the weakest, Qwen2.5-VL-3B, is barely above the 0.288 no-pixel floor: naming a statistical finding from a chart is not something current general-purpose VLMs do reliably without task-specific training.

Rendering matters as much as data. Trained on matplotlib alone, models lose 8.5–10.6 points on Plotly re-renders of the same data and 22–25 points on Vega-Lite. Adding the supplementary real pool changes real accuracy by less than a point for ResNet-101, but training-only renderer augmentation closes most of the Plotly gap (a renderer it has seen) and, more tellingly, lifts accuracy on Vega-Lite, which no model has seen, from 0.60–0.64 to 0.65–0.71 without costing matplotlib accuracy. A residual 15-point unseen-renderer gap remains and is the main open problem the dataset exposes.

Errors concentrate in the distribution family (0.763 on real charts, versus 0.886–0.930 for the other families) and in classes with few real examples: heavy tails (0.385, $n = 26$), seasonality (0.462, $n = 13$) and left skew (0.474, $n = 19$). The commonest confusions are between classes that differ in degree rather than kind: bimodal read as right-skewed (19 of 100), heavy-tailed read as right-skewed (13 of 26), and significant group differences read as none (12 of 100). Macro-F1 on real charts is 0.776. Per-class results for the rarest real classes are indicative only: with $n = 13$ to 26 real test charts, a class's 95% interval spans roughly ±19 to ±27 points.

Usage Notes

- Report real-only accuracy. The corpus is 70% synthetic, and synthetic charts are easier; mixed accuracy hides the generalisation gap. Use splits/test_real_only.jsonl.
- Report an unseen renderer. Score the Vega-Lite re-renders of the real test charts, and do not train on Vega-Lite. If the Plotly training renders are used for augmentation, the Plotly test renders measure a seen renderer and should be labelled so.
- Report against the floor, not 5%. Chart type alone gives 0.288.
- Also report the deduplicated test set. near_duplicates.json lists 85 real test charts whose data recurs in training data under another source table.
- Do not flip images. Horizontal or vertical flips change the label for roughly 30% of samples.

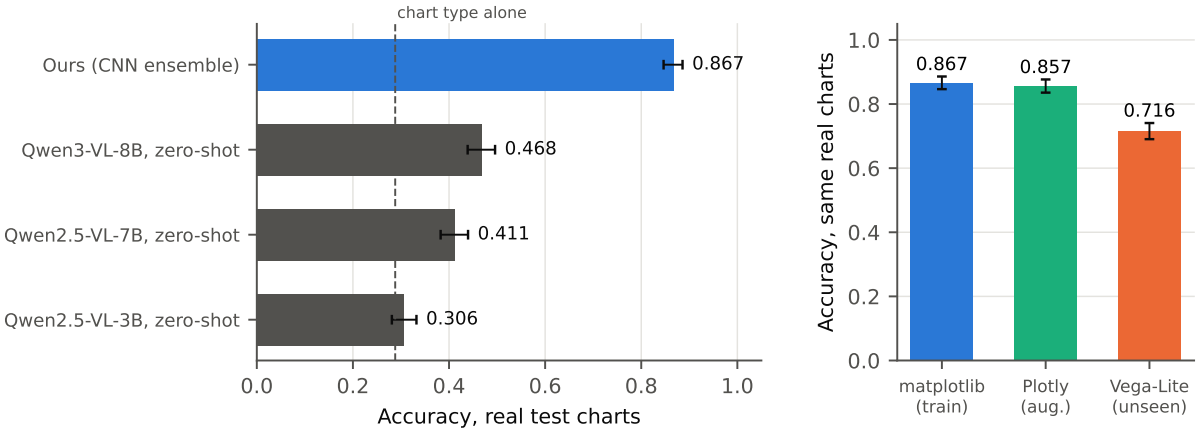


Figure 3: Left: accuracy on the 1,192 real test charts (matplotlib renders), 95% bootstrap CIs; dashed line, the chart-type prior. Right: the deployed ensemble on the same real charts drawn by the training renderer, by Plotly (used only as training augmentation) and by Vega-Lite (never seen).

- Tune on validation only. Real validation has 553 charts; selecting models or ensembles on the test split inflates results (an earlier draft of this work did so).
- Single label. Charts showing two findings at once were excluded, so models trained here will not report compound findings.
- Thresholds are conventions. $|r| > 0.7$ for "strong" or $\varepsilon^2 > 0.14$ for "large" follow common practice; every guard is recorded in class_index.json and the verification code, so labels can be re-derived under other choices.
- Class priors are artificial. Balance is a training convenience; real-world finding frequencies are skewed. The real subsets reflect public-data frequencies.

Code Availability

The complete pipeline (harvesting, mining, synthesis, the verification battery, rendering, validation, export, the shortcut audit, the Plotly renderer, training notebooks, ensembling, statistics and figures) is available at [repository URL]. The corpus rebuilds end to end in under an hour on a laptop CPU.

Author Contributions

[Contributions per author.]

Competing Interests

The authors declare no competing interests.

Supplementary Information

Supplementary Table S1. Real-source samples per class and split, and in the supplementary training pool (train-split sources only). Synthetic samples fill each class to 700/100/200 in the balanced splits.

Class	Train	Val	Test	Supplementary
dist_normal	126	16	21	–
dist_right_skew	350	50	100	208
dist_left_skew	48	8	19	–
dist_bimodal	350	50	100	400
dist_heavy_tail	116	21	26	–
dist_uniform	74	19	21	–
rel_strong_positive	240	42	84	–
rel_strong_negative	45	9	14	–
rel_none	350	50	100	400
rel_nonlinear_monotonic	118	29	42	–
rel_nonlinear_nonmonotonic	6	1	2	–
rel_heteroscedastic	350	50	100	143
ts_increasing	350	50	100	400
ts_decreasing	350	8	100	140
ts_seasonal	47	4	13	–
ts_changepoint	145	4	26	–
ts_anomaly_spike	350	5	66	237
grp_significant_diff	350	50	100	245
grp_no_diff	350	50	100	372
grp_dominant_category	223	37	58	–
Total	4,338	553	1,192	2,545

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